Continuum Mechanics — Calculus Edition

Field 22 of 80 — The LKVexa Compendium carried to the continuum

Contents

Continuum Mechanics — Calculus Image **1**
Kinematics of Deformation 1
Stress and Balance Laws 13
Linear Elasticity 26
Nonlinear and Inelastic Behaviour 40

Continuum Mechanics — Calculus Image

This field is written in matrices already: the deformation gradient, the strain and stress tensors, and the elastic stiffness are all matrices, and the central results are their decompositions — polar, spectral, volumetric–deviatoric. Small strain is exactly linear; finite deformation is not, and the distinction is carried honestly in every *Status* line of the kinematics section.

Each entry carries its originating equation from the Linear Algebra edition, its 8S image, its calculus image under $\mathsf{T}_{\mathrm{cal}}$, a six-step proof, and the original discussion. The schema (Q1)–(Q6), the master theorems Θ-A…Θ-J and the lemmas D1…D14 are established in `00_Foundations_Calculus_Transform.md`.

Kinematics of Deformation

1. Deformation Gradient

Origin (LA edition).

$$F = \frac{\partial \mathbf{x}}{\partial \mathbf{X}}, \qquad F_{iJ} = \frac{\partial x_i}{\partial X_J}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(1)} = \mu_k^{(1)}\phi_k^{(1)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(1)} \le \mu_2^{(1)} \le \cdots \le \mu_N^{(1)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(1)}) = \mu_k^{(1)}\phi_k^{(1)}, \qquad \mu_2^{(1)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_1 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(1)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_1$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The Jacobian of the deformation map.

Structure: The single object from which all strain measures are built. F is not symmetric and includes rotation; $\det F > 0$ enforces non-inversion of material. Every kinematic quantity in this chapter is a function of F.

Reduction: No reduction needed — F is a matrix by definition.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a first-order operator, which the transform leaves in place.

2. Jacobian of Deformation

Origin (LA edition).

$$J = \det F$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(2)} = \mu_k^{(2)} \phi_k^{(2)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(2)} \leq \mu_2^{(2)} \leq \cdots \leq \mu_N^{(2)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(2)}) = \mu_k^{(2)} \phi_k^{(2)}, \qquad \mu_2^{(2)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_2 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(2)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_2$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The local volume ratio.

Structure: $J = 1$ is incompressibility, $J > 0$ is required for physical admissibility. Its derivative is $\dot{J} = J \operatorname{tr}(L) = J \nabla \cdot \mathbf{v}$ by Jacobi's formula — the continuity equation in disguise.

Reduction: Volume change is a determinant.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

3. Polar Decomposition

Origin (LA edition).

$$F = RU = VR, \qquad R^\top R = I,\ U = U^\top \succ 0$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(3)} = \mu_k^{(3)} \phi_k^{(3)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(3)} \leq \mu_2^{(3)} \leq \cdots \leq \mu_N^{(3)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(3)}) = \mu_k^{(3)} \phi_k^{(3)}, \qquad \mu_2^{(3)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_3 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum

could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(3)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_3$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Rotation times stretch.

Structure: Separates rigid rotation from pure deformation uniquely. $U = (F^\top F)^{1/2}$ is the right stretch tensor with principal stretches as eigenvalues; R is the nearest orthogonal matrix to F in every unitarily invariant norm. This is the same decomposition as the matrix polar decomposition of field 05.

Reduction: Already matrix-native.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Linear Algebra.

4. Right Cauchy–Green Tensor

Origin (LA edition).

$$C = F^\top F$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(4)} = \mu_k^{(4)} \phi_k^{(4)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(4)} \leq \mu_2^{(4)} \leq \cdots \leq \mu_N^{(4)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(4)}) = \mu_k^{(4)} \phi_k^{(4)}, \qquad \mu_2^{(4)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_4 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum

could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(4)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_4$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A symmetric positive definite Gram matrix.

Structure: Rotation-invariant, since R cancels: C depends only on the stretch. Its eigenvalues are the squared principal stretches λ_i^2, and $\det C = J^2$. It is the metric induced by the deformation.

Reduction: The tensor is a Gram matrix of F.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

5. Green–Lagrange Strain

Origin (LA edition).

$$E = \tfrac{1}{2}(C - I) = \tfrac{1}{2}(F^\top F - I)$$

Input (8S edition).

$$\max_{n_5 \in \mathbb{Z}^5} \; \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_5\|_1 \leq M, \;\; \eta^{(5)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_5\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(5)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_5\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(5)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_5 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality

constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(5)}_{\mathrm{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_5$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A symmetric strain measure, zero for rigid motion.

Structure: Quadratic in F, hence nonlinear in the displacement — this is exactly where geometric nonlinearity enters. Expanding in the displacement gradient H gives $E = \frac{1}{2}(H + H^\top) + \frac{1}{2}H^\top H$; dropping the quadratic term gives small strain.

Reduction: Quadratic in the deformation gradient.

Status: Linearised to give the infinitesimal strain.

8S status: Linearised to give the infinitesimal strain; the 8S image inherits the same approximation status (Theorem C, Lemma T11).

Calculus status: Linearised to give the infinitesimal strain; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

6. Infinitesimal Strain Tensor

Origin (LA edition).

$$\varepsilon = \tfrac{1}{2}(H + H^\top) = \mathrm{sym}(H), \qquad H = \nabla \mathbf{u}$$

Input (8S edition).

$$\left(g^{(5)}_{6j},\ \delta^{(8)}_{6j},\ g^{(3)}_{6j},\ g^{(\mathrm{J})}_{6j}\right) \ \longmapsto \ \mathcal{R}^{8S}_{6j}, \qquad g^{(3)}_{6j} \leq g^{(5)}_{6j}$$

Calculus image.

$$d_g(z_6, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{6j} \leq d_g(z_6, z_j), \qquad g^{(3)}_{6j} \leq g^{(5)}_{6j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_6 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \leq d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{6j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(6)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_6$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The symmetric part of the displacement gradient.

Structure: $H = \operatorname{sym}(H) + \operatorname{skew}(H)$ splits deformation from infinitesimal rotation exactly, and the skew part $\operatorname{skew}(H) = \frac{1}{2}[\omega]_\times$ is the rotation vector. Valid when $\|H\| \ll 1$; the whole of linear elasticity rests on this truncation.

Reduction: Strain is the symmetric part of a matrix.

Status: Exact linear (by construction).

8S status: Exact linear (by construction); preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear (by construction); carried to the continuum by Lemma D9, and the input already carried a first-order operator, which the transform leaves in place.

7. Volumetric and Deviatoric Split

Origin (LA edition).

$$\varepsilon = \tfrac{1}{3}\operatorname{tr}(\varepsilon) I + \varepsilon', \qquad \operatorname{tr}\varepsilon' = 0$$

Input (8S edition).

$$P_5^{(7)} = C_5 C_5^{+} = P_4 + \frac{w_7 w_7^{\top}}{\|w_7\|_2^2}, \qquad w_7 = (I - C_4 C_4^{+}) e_7, \qquad \eta^{(7)}_{\text{ind}} = \frac{\|w_7\|_2^2}{\|e_7\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_7 - P_{\mathcal{V}} e_7,\ v \rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta^{(7)}_{\text{ind}} = \frac{\det \operatorname{Gr}(C_4, e_7)}{\det \operatorname{Gr}(C_4)(\|e_7\|^2 + \varepsilon)}, \qquad \nabla_{e_7} \|w_7\|^2 = 2\, w_7$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_7 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\operatorname{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_7\|^2/(\|e_7\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and

$\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(7)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_7$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Orthogonal decomposition into trace and traceless parts.

Structure: An orthogonal projection in the Frobenius inner product: the spherical and deviatoric subspaces of symmetric matrices are orthogonal, of dimensions 1 and 5 in three dimensions. Volume change and shape change decouple, which is why bulk and shear moduli are independent.

Reduction: The split is an orthogonal projection of matrix space.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

8. Compatibility Condition

Origin (LA edition).

$$\nabla \times \big(\nabla \times \varepsilon\big)^{\top} = 0$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta^{(8)}_{\text{ind}} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_8) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_8 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(8)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_8$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Integrability of a strain field to a displacement field.

Structure: $\varepsilon \in \mathrm{im}(\mathrm{sym} \circ \nabla)$ exactly when the curl-curl vanishes — a cohomological condition. Six strain components come from three displacements, so there are three independent compatibility equations; the count is a rank–nullity statement.

Reduction: Compatibility is a column-space membership.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input already carried a first-order operator, which the transform leaves in place.

9. Velocity Gradient

Origin (LA edition).

$$L = \frac{\partial \mathbf{v}}{\partial \mathbf{x}} = \dot{F} F^{-1}$$

Input (8S edition).

$$\left(g_{9j}^{(5)},\ \delta_{9j}^{(8)},\ g_{9j}^{(3)},\ g_{9j}^{(\mathrm{J})}\right) \ \longmapsto\ \mathcal{R}_{9j}^{8S}, \qquad g_{9j}^{(3)} \le g_{9j}^{(5)}$$

Calculus image.

$$d_g(z_9, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta_{9j}^{(8)} \le d_g(z_9, z_j), \qquad g_{9j}^{(3)} \le g_{9j}^{(5)}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_9 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta_{9j}^{(8)}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(9)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_9$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The spatial rate of deformation matrix.

Structure: $\operatorname{tr} L = \nabla \cdot \mathbf{v}$ is the volumetric rate; its symmetric and skew parts separate stretching from spin. The identity $L = \dot{F}F^{-1}$ is the continuum analogue of the logarithmic derivative.

Reduction: Already matrix-native.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input already carried a first-order operator, which the transform leaves in place.

10. Rate of Deformation and Spin

Origin (LA edition).

$$D = \operatorname{sym}(L), \qquad W = \operatorname{skew}(L)$$

Input (8S edition).

$$\max_{n_{10} \in \mathbb{Z}^5} \ \Delta_{8S} = \operatorname{Score}(\mathcal{M}_8) - \operatorname{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{10}\|_1 \le M, \ \ \eta_{\text{ind}}^{(10)} \ge \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{10}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(10)})\Big] = 0, \qquad \lambda, \nu \ge 0, \qquad \lambda(\|n_{10}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(10)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{10} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \ge 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(10)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{10}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The additive split of the velocity gradient.

Structure: D drives dissipation ($\sigma : D$ is the stress power) while W is the rigid spin and does no work against a symmetric stress. This is why objective stress rates subtract W — the Jaumann rate $\mathring{\sigma} = \dot{\sigma} - W\sigma + \sigma W$.

Reduction: The split is the symmetric–skew decomposition.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

11. Material (Total) Derivative

Origin (LA edition).

$$\frac{D\phi}{Dt} = \frac{\partial \phi}{\partial t} + \mathbf{v}^\top \nabla \phi$$

Input (8S edition).

$$E_{11}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 11}, \qquad \|z_{11}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta_{11j}^{(8)} = \sqrt{(c_{11}-c_j)^\top G_5 (c_{11}-c_j) + (r_{11}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{11}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{11} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{11}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{11}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{11} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(11)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{11}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A linear operator on fields, discretised as $\partial_t + VD$.

Structure: Linear in ϕ for a *given* velocity field, and it is the velocity-dependence that makes Navier–Stokes nonlinear. The discrete advection operator VD is skew-symmetric for divergence-free $\mathbf{v}$ on a periodic grid, hence energy-conserving.

Reduction: The derivative is a linear operator for fixed $\mathbf{v}$.

Status: Exact linear in the transported field.

8S status: Exact linear in the transported field; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear in the transported field; carried to the continuum by Lemma D3, and the input already carried a first-order operator, which the transform leaves in place.

12. Reynolds Transport Theorem

Origin (LA edition).

$$\frac{d}{dt}\int_{\Omega(t)} \phi \, dV = \int_{\Omega} \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}) \right) dV$$

Input (8S edition).

$$\left(g^{(5)}_{12j},\ \delta^{(8)}_{12j},\ g^{(3)}_{12j},\ g^{(\mathrm{J})}_{12j} \right) \ \longmapsto\ \mathcal{R}^{8S}_{12j}, \qquad g^{(3)}_{12j} \le g^{(5)}_{12j}$$

Calculus image.

$$d_g(z_{12}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{12j} \le d_g(z_{12}, z_j), \qquad g^{(3)}_{12j} \le g^{(5)}_{12j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^{\top} G_5^{-1} \xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{12} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{12j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(12)}_{\mathrm{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{12}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Differentiation of an integral over a moving domain.

Structure: Derived from Jacobi's formula $\dot{J} = J \operatorname{tr} L$ — the domain motion contributes exactly the divergence term. Every conservation law in the field follows by applying this theorem to a different ϕ.

Reduction: The theorem follows from the determinant derivative.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

Stress and Balance Laws

13. Cauchy Stress Tensor

Origin (LA edition).

$$\mathbf{t} = \sigma \mathbf{n}$$

Input (8S edition).

$$\left(g^{(5)}_{13j},\ \delta^{(8)}_{13j},\ g^{(3)}_{13j},\ g^{(\mathrm{J})}_{13j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{13j}, \qquad g^{(3)}_{13j} \le g^{(5)}_{13j}$$

Calculus image.

$$d_g(z_{13}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{13j} \le d_g(z_{13}, z_j), \qquad g^{(3)}_{13j} \le g^{(5)}_{13j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{13} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{13j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(13)}_{\mathrm{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{13}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A matrix mapping surface normals to traction vectors.

Structure: Cauchy's theorem asserts that traction depends linearly on the normal — the existence of the stress tensor is precisely a linearity claim, proved by a tetrahedron argument. It is the founding linear structure of the field.

Reduction: Traction is a matrix–vector product.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

14. Symmetry of Cauchy Stress

Origin (LA edition).

$$\sigma = \sigma^{\top}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(14)} = \mu_k^{(14)}\phi_k^{(14)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(14)} \le \mu_2^{(14)} \le \cdots \le \mu_N^{(14)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(14)}) = \mu_k^{(14)}\phi_k^{(14)}, \qquad \mu_2^{(14)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{14} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^{\top}\mathcal{L}_{8S}x/x^{\top}x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(14)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{14}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Symmetry forced by angular momentum balance.

Structure: Reduces nine components to six and guarantees real eigenvalues with orthogonal eigenvectors — the spectral theorem applies, giving principal stresses and directions. Symmetry fails only in micropolar media with body couples.

Reduction: The balance law forces matrix symmetry.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

15. First Piola–Kirchhoff Stress

Origin (LA edition).

$$P = J\sigma F^{-\top}$$

Input (8S edition).

$$\left(g^{(5)}_{15j},\ \delta^{(8)}_{15j},\ g^{(3)}_{15j},\ g^{(\mathrm{J})}_{15j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{15j}, \qquad g^{(3)}_{15j} \le g^{(5)}_{15j}$$

Calculus image.

$$d_g(z_{15}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot\Gamma\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{15j} \le d_g(z_{15}, z_j), \qquad g^{(3)}_{15j} \le g^{(5)}_{15j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{15} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{15j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(15)}_{\mathrm{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{15}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A two-point tensor, generally non-symmetric.

Structure: Relates forces in the current configuration to areas in the reference one, which is what makes it the natural conjugate to $\dot F$: the stress power is $P : \dot F$. Non-symmetry reflects its mixed-configuration character.

Reduction: A matrix congruence-like transformation of σ.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

16. Second Piola–Kirchhoff Stress

Origin (LA edition).

$$S = JF^{-1}\sigma F^{-\top} = F^{-1}P$$

Input (8S edition).

$$\max_{n_{16}\in\mathbb{Z}^5} \Delta_{8S} = \mathrm{Score}(\mathcal{M}_8) - \mathrm{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{16}\|_1 \le M, \;\; \eta_{\mathrm{ind}}^{(16)} \ge \varepsilon_{\mathrm{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{16}\|_1 - M) - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(16)})\Big] = 0, \qquad \lambda,\nu \ge 0, \qquad \lambda(\|n_{16}\|_1 - M) = \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(16)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{16} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \ge 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(16)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{16}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A fully referential symmetric stress.

Structure: Symmetric and work-conjugate to the Green–Lagrange strain: $S : \dot{E}$ is the stress power. Being a full congruence transformation $F^{-1}(\cdot)F^{-\top}$, it preserves symmetry and inertia — the signature of the stress state is frame-independent.

Reduction: The transformation is a matrix congruence.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

17. Conservation of Mass (Continuity)

Origin (LA edition).

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \qquad \rho J = \rho_0$$

Input (8S edition).

$$\Big(g_{17j}^{(5)},\; \delta_{17j}^{(8)},\; g_{17j}^{(3)},\; g_{17j}^{(\mathrm{J})}\Big) \;\longmapsto\; \mathcal{R}_{17j}^{8S}, \qquad g_{17j}^{(3)} \le g_{17j}^{(5)}$$

Calculus image.

$$d_g(z_{17}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{17j} \le d_g(z_{17}, z_j), \qquad g^{(3)}_{17j} \le g^{(5)}_{17j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{17} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{17j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(17)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{17}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A divergence-form conservation law; discretely $\dot{\rho} = -D(\rho \odot \mathbf{v})$.

Structure: Linear in ρ for a given velocity field. The referential form $\rho J = \rho_0$ is a determinant statement — mass conservation is conservation of $\rho \det F$.

Reduction: Continuity is linear in the density.

Status: Exact linear in ρ.

8S status: Exact linear in ρ; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear in ρ; carried to the continuum by Lemma D9, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

18. Cauchy's Equation of Motion

Origin (LA edition).

$$\rho \frac{D\mathbf{v}}{Dt} = \nabla \cdot \sigma + \rho \mathbf{b}$$

Input (8S edition).

$$\left(g^{(5)}_{18j},\ \delta^{(8)}_{18j},\ g^{(3)}_{18j},\ g^{(\mathrm{J})}_{18j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{18j}, \qquad g^{(3)}_{18j} \le g^{(5)}_{18j}$$

Calculus image.

$$d_g(z_{18}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{18j} \le d_g(z_{18}, z_j), \qquad g^{(3)}_{18j} \le g^{(5)}_{18j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{18} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{18j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(18)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{18}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Divergence of a matrix field.

Structure: $\nabla \cdot \sigma$ is a linear operator on the stress field, discretised as $B^\top \sigma$ with B the discrete gradient. The left side is nonlinear only through the convective term $\mathbf{v}^\top \nabla \mathbf{v}$.

Reduction: The divergence is a linear operator on the stress.

Status: Exact linear apart from convection.

8S status: Exact linear apart from convection; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear apart from convection; carried to the continuum by Lemma D9, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

19. Static Equilibrium

Origin (LA edition).

$$\nabla \cdot \sigma + \rho \mathbf{b} = \mathbf{0}$$

Input (8S edition).

$$W_{19j} = A_{19j} \exp\left[-\frac{(c_{19} - c_j)^\top G_5 (c_{19} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(19, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 19} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{19} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2}x$, under which $\nabla\cdot(G_5^{-1}\nabla\,\cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2}e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|^2_{G_5}/2\sigma_c^2 = \|x\|^2_{G_5}/4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top Ax}e^{-i\xi\cdot x}dx = \pi^{5/2}\det(A)^{-1/2}e^{-\xi^\top A^{-1}\xi/4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(19)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{19}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Discretised, $K\mathbf{u} = \mathbf{f}$.

Structure: With linear elasticity the assembled system is symmetric positive definite once rigid-body modes are constrained — the null space of K is exactly the six rigid-body motions in three dimensions. Solving structures is solving this system.

Reduction: Equilibrium becomes a sparse SPD linear system.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

Also in: Structural Engineering.

20. Energy Balance

Origin (LA edition).

$$\rho\dot{e} = \sigma : D - \nabla\cdot\mathbf{q} + \rho r$$

Input (8S edition).

$$E_{20}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node }20}, \qquad \big\|z_{20}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta^{(8)}_{20j} = \sqrt{(c_{20}-c_j)^\top G_5(c_{20}-c_j) + (r_{20}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{20}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{20} = 0 \iff \mathcal{L}_\star\phi = 0, \qquad ds^2_{20} = dc^\top(G_5 + \nabla r\,\nabla r^\top)dc + r_{20}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{20} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(20)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{20}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The stress power as a Frobenius inner product.

Structure: $\sigma : D = \operatorname{tr}(\sigma^\top D)$ is bilinear in stress and rate of deformation. Only the symmetric part D contributes, since σ is symmetric and the trace of a symmetric times a skew matrix vanishes.

Reduction: Stress power is a matrix inner product.

Status: Exact linear (bilinear).

8S status: Exact linear (bilinear); preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear (bilinear); carried to the continuum by Lemma D3, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

21. Clausius–Duhem Inequality

Origin (LA edition).

$$\rho\dot{\eta} + \nabla\cdot\left(\frac{\mathbf{q}}{T}\right) - \frac{\rho r}{T} \geq 0$$

Input (8S edition).

$$\text{Admit}_{21}(8S) \iff \eta^{(21)}_{\text{ind}} \geq \varepsilon_{\text{ind}} \;\wedge\; \Delta_{8S} > \varepsilon_{\text{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \;\text{ or }\; \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{21}(x,\lambda) = 0, \qquad \partial_x\mathcal{G}_{21} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x\mathcal{G}_{21} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore

does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{21} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x, \lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(21)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{21}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A sign constraint on a bilinear dissipation form.

Structure: Restricts admissible constitutive laws: for viscous materials it forces the viscosity tensor to be positive semidefinite, since dissipation $D : \mathbb{C} : D \geq 0$ must hold for all D. Thermodynamic admissibility is a definiteness condition.

Reduction: The second law becomes positive semidefiniteness of a material tensor.

Status: Exact linear as a definiteness condition.

8S status: Exact linear as a definiteness condition; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear as a definiteness condition; carried to the continuum by Lemma D7, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

22. Principal Stresses

Origin (LA edition).

$$(\sigma - \sigma I)\mathbf{n} = \mathbf{0}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(22)} = \mu_k^{(22)}\phi_k^{(22)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(22)} \leq \mu_2^{(22)} \leq \cdots \leq \mu_N^{(22)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(22)}) = \mu_k^{(22)}\phi_k^{(22)}, \qquad \mu_2^{(22)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{22} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(22)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{22}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The eigenproblem of the stress tensor.

Structure: Symmetry gives three real eigenvalues and orthogonal principal directions. On those planes the traction is purely normal — shear vanishes. Failure criteria are almost always stated in terms of these eigenvalues.

Reduction: Principal stresses are eigenvalues.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

23. Stress Invariants

Origin (LA edition).

$$I_1 = \operatorname{tr}\sigma, \quad I_2 = \tfrac{1}{2}\big[(\operatorname{tr}\sigma)^2 - \operatorname{tr}\sigma^2\big], \quad I_3 = \det\sigma$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(23)}_k = \mu^{(23)}_k \phi^{(23)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(23)}_1 \leq \mu^{(23)}_2 \leq \cdots \leq \mu^{(23)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(23)}_k) = \mu^{(23)}_k \phi^{(23)}_k, \qquad \mu^{(23)}_2 = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{23} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(23)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{23}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Coefficients of the characteristic polynomial.

Structure: Invariant under rotation because they are the elementary symmetric functions of the eigenvalues. Any isotropic scalar function of stress must depend on these three alone — the representation theorem for isotropic functions.

Reduction: The invariants are characteristic polynomial coefficients.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

24. Von Mises Equivalent Stress

Origin (LA edition).

$$\sigma_v = \sqrt{\tfrac{3}{2}\,\sigma' : \sigma'} = \sqrt{\tfrac{3}{2}}\,\|\sigma'\|_F$$

Input (8S edition).

$$\left(g^{(5)}_{24j},\ \delta^{(8)}_{24j},\ g^{(3)}_{24j},\ g^{(\mathrm{J})}_{24j}\right) \;\longmapsto\; \mathcal{R}^{8S}_{24j}, \qquad g^{(3)}_{24j} \leq g^{(5)}_{24j}$$

Calculus image.

$$d_g(z_{24}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{24j} \leq d_g(z_{24}, z_j), \qquad g^{(3)}_{24j} \leq g^{(5)}_{24j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{24} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{24j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(24)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{24}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The Frobenius norm of the deviatoric stress.

Structure: Depends only on the deviator, hence pressure-independent — appropriate for metals, whose yielding is volume-preserving. The yield surface $\sigma_v = \sigma_y$ is a cylinder in principal stress space with axis along $\mathbf{1}$.

Reduction: The equivalent stress is a matrix norm.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

25. Mohr's Circle Relations

Origin (LA edition).

$$\sigma_n = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2}\cos 2\theta$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(25)}_k = \mu^{(25)}_k \phi^{(25)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(25)}_1 \le \mu^{(25)}_2 \le \cdots \le \mu^{(25)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(25)}_k) = \mu^{(25)}_k \phi^{(25)}_k, \qquad \mu^{(25)}_2 = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{25} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum

could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(25)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{25}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A graphical representation of a 2×2 eigenvalue problem.

Structure: The circle is the locus of (σ_n, τ) as the plane rotates — a parametrisation of the orbit of the stress tensor under $SO(2)$ conjugation. Centre is $\frac{1}{2}\,\mathrm{tr}$, radius is half the eigenvalue spread.

Reduction: The circle is the rotation orbit of a symmetric matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

26. Maximum Shear Stress

Origin (LA edition).

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(26)} = \mu_k^{(26)} \phi_k^{(26)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(26)} \leq \mu_2^{(26)} \leq \cdots \leq \mu_N^{(26)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(26)}) = \mu_k^{(26)} \phi_k^{(26)}, \qquad \mu_2^{(26)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{26} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(26)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{26}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Half the eigenvalue spread.

Structure: Half the difference between the largest and smallest eigenvalues, attained on planes at 45° to the corresponding principal directions. It is $\frac{1}{2}(\lambda_{\max} - \lambda_{\min})$ — a spectral spread, and the basis of the Tresca criterion.

Reduction: Maximum shear is a spectral spread.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Linear Elasticity

27. Generalised Hooke's Law

Origin (LA edition).

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}$$

Input (8S edition).

$$W_{27j} = A_{27j} \exp\left[-\frac{(c_{27} - c_j)^\top G_5 (c_{27} - c_j)}{2\sigma_c^2} - \frac{d_{\text{J}}(27, j)^2}{2\sigma_{\text{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \;\; k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 27} = e^{t\mathcal{A}_G}, \;\; t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\, \xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{27} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2}x$, under which $\nabla\cdot(G_5^{-1}\nabla\,\cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2}e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|^2_{G_5}/2\sigma_c^2 = \|x\|^2_{G_5}/4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top Ax}e^{-i\xi\cdot x}dx = \pi^{5/2}\det(A)^{-1/2}e^{-\xi^\top A^{-1}\xi/4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(27)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{27}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A fourth-order stiffness tensor; a 6×6 matrix in Voigt notation.

Structure: Symmetries reduce 81 components to 21 for the most general anisotropic material: minor symmetries from stress and strain symmetry, major symmetry from the existence of a strain energy. In Voigt form C is a 6×6 symmetric positive definite matrix.

Reduction: The fourth-order tensor becomes a 6×6 matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

28. Isotropic Hooke's Law

Origin (LA edition).

$$\sigma = \lambda\,\mathrm{tr}(\varepsilon)I + 2\mu\varepsilon$$

Input (8S edition).

$$P_5^{(28)} = C_5C_5^+ = P_4 + \frac{w_{28}w_{28}^\top}{\|w_{28}\|_2^2}, \qquad w_{28} = (I - C_4C_4^+)e_{28}, \qquad \eta^{(28)}_{\text{ind}} = \frac{\|w_{28}\|_2^2}{\|e_{28}\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_{28} - P_{\mathcal{V}}e_{28},\ v\rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta^{(28)}_{\text{ind}} = \frac{\det \mathrm{Gr}(C_4, e_{28})}{\det \mathrm{Gr}(C_4)(\|e_{28}\|^2 + \varepsilon)}, \qquad \nabla_{e_{28}}\|w_{28}\|^2 = 2\,w_{28}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_{28} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram

matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_{28}\|^2/(\|e_{28}\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(28)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{28}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Two constants suffice under full isotropy.

Structure: The stiffness is $C = 3K\,\mathbb{P}_{\text{vol}} + 2\mu\,\mathbb{P}_{\text{dev}}$ — a spectral decomposition into the two orthogonal projections of the volumetric–deviatoric split, with eigenvalues $3K$ (multiplicity 1) and 2μ (multiplicity 5). Isotropic elasticity has exactly two distinct eigenvalues.

Reduction: The law is a spectral decomposition of the stiffness.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

29. Elastic Modulus Relations

Origin (LA edition).

$$E = \frac{9K\mu}{3K + \mu}, \qquad \nu = \frac{3K - 2\mu}{2(3K + \mu)}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(29)} = \mu_k^{(29)}\phi_k^{(29)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(29)} \leq \mu_2^{(29)} \leq \cdots \leq \mu_N^{(29)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(29)}) = \mu_k^{(29)}\phi_k^{(29)}, \qquad \mu_2^{(29)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{29} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue

equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(29)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{29}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Reparametrisations of the same two eigenvalues.

Structure: All modulus pairs (E, ν), (K, μ), (λ, μ) describe the same two eigenvalues of C. Positive definiteness requires $K > 0$ and $\mu > 0$, which translates to $-1 < \nu < 1/2$ — the physical range follows from a definiteness condition.

Reduction: The relations reparametrise two eigenvalues.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

30. Uniaxial Hooke's Law

Origin (LA edition).

$$\sigma = E\varepsilon$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(30)} = \mu_k^{(30)} \phi_k^{(30)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(30)} \leq \mu_2^{(30)} \leq \cdots \leq \mu_N^{(30)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(30)}) = \mu_k^{(30)} \phi_k^{(30)}, \qquad \mu_2^{(30)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{30} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue

equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(30)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{30}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The 1×1 case.

Structure: The scalar law recovered by restricting the tensorial law to a uniaxial stress state and eliminating the lateral strains. E is not an eigenvalue of C — it is a compliance-derived quantity, which is why E and μ are not directly comparable.

Reduction: The scalar law is a restriction of the tensorial one.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

31. Poisson's Ratio

Origin (LA edition).

$$\nu = -\frac{\varepsilon_{\text{lat}}}{\varepsilon_{\text{axial}}}$$

Input (8S edition).

$$W_{31j} = A_{31j} \exp\left[-\frac{(c_{31} - c_j)^\top G_5 (c_{31} - c_j)}{2\sigma_c^2} - \frac{d_{\text{J}}(31, j)^2}{2\sigma_{\text{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 31} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{31} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|_{G_5}^2 / 2\sigma_c^2 = \|x\|_{G_5}^2 / 4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi / 4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(31)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{31}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An off-diagonal entry of the compliance matrix.

Structure: Bounded by $-1 < \nu < 1/2$ from positive definiteness of C; $\nu = 1/2$ is exactly incompressibility ($K \to \infty$), where the stiffness matrix becomes singular in the volumetric direction — the source of locking in finite elements.

Reduction: The ratio is a compliance matrix entry.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

32. Navier–Cauchy Equations

Origin (LA edition).

$$(\lambda + \mu)\nabla(\nabla \cdot \mathbf{u}) + \mu\nabla^2\mathbf{u} + \rho\mathbf{b} = \mathbf{0}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(32)} = \mu_k^{(32)}\phi_k^{(32)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(32)} \leq \mu_2^{(32)} \leq \cdots \leq \mu_N^{(32)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(32)}) = \mu_k^{(32)}\phi_k^{(32)}, \qquad \mu_2^{(32)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{32} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(32)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{32}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A vector elliptic operator, discretised as a symmetric positive definite K.

Structure: The operator is $\mu\Delta + (\lambda+\mu)\nabla\nabla\cdot$, which the Helmholtz decomposition diagonalises into curl-free and divergence-free parts with distinct eigenvalues — giving the P and S wave speeds directly.

Reduction: The PDE becomes a sparse SPD system.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

33. P-Wave and S-Wave Speeds

Origin (LA edition).

$$c_p = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \qquad c_s = \sqrt{\frac{\mu}{\rho}}$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi^{(33)}_k = \mu^{(33)}_k \phi^{(33)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(33)}_1 \leq \mu^{(33)}_2 \leq \cdots \leq \mu^{(33)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(33)}_k) = \mu^{(33)}_k \phi^{(33)}_k, \qquad \mu^{(33)}_2 = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{33} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(33)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{33}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Eigenvalues of the acoustic (Christoffel) tensor.

Structure: The Christoffel tensor $\Gamma_{ik} = C_{ijkl} n_j n_l$ is symmetric positive definite, and its three eigenvalues divided by ρ give the squared wave speeds — one longitudinal, two transverse. Anisotropy splits the two shear speeds.

Reduction: Wave speeds are eigenvalues of the Christoffel matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Geophysics, Acoustics.

34. Strain Energy Density

Origin (LA edition).

$$W = \tfrac{1}{2} \varepsilon : C : \varepsilon$$

Input (8S edition).

$$W_{34j} = A_{34j} \exp\left[-\frac{(c_{34} - c_j)^\top G_5 (c_{34} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(34, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 34} = e^{t \mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\, \xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{34} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|^2_{G_5}/2\sigma_c^2 = \|x\|^2_{G_5}/4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi / 4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition,

which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(34)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{34}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A quadratic form in the strain.

Structure: Positive definiteness of C is exactly the requirement that deformation costs energy — material stability. Its Hessian with respect to strain *is* C, which is why the stiffness is symmetric (equality of mixed partials).

Reduction: Strain energy is a quadratic form.

Status: Exact linear (quadratic form).

8S status: Exact linear (quadratic form); preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear (quadratic form); carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

35. Airy Stress Function

Origin (LA edition).

$$\sigma_{xx} = \partial_y^2 \phi, \quad \sigma_{yy} = \partial_x^2 \phi, \quad \sigma_{xy} = -\partial_x \partial_y \phi$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta^{(35)}_{\text{ind}} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_{35}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{35} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(35)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{35}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A potential automatically satisfying equilibrium.

Structure: Equilibrium is satisfied identically because the construction places σ in $\ker(\nabla\cdot)$ — a null-space parametrisation. Compatibility then requires $\nabla^4\phi = 0$, the biharmonic equation.

Reduction: The function parametrises the null space of the divergence.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

36. Stress Concentration Factor

Origin (LA edition).

$$K_t = \frac{\sigma_{\max}}{\sigma_{\text{nom}}}$$

Input (8S edition).

$$y_{36}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{36})\Big]z(u), \qquad z_{36}(u) = \begin{bmatrix} c_{36} \\ r_{36}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{36})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{36}\,\psi_{36}(c')\,p(c')\,dc', \qquad \partial_\tau\psi_{36} = -\mathcal{L}_\star\psi_{36}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{36} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{36})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(36)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{36}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An amplification factor, structurally a condition number.

Structure: For an elliptical hole $K_t = 1 + 2a/b$, diverging as the ellipse sharpens into a crack. The amplification of a local field by geometry is exactly the role a condition number plays for a linear system.

Reduction: The factor is a field amplification ratio.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

37. Stress Intensity Factor

Origin (LA edition).

$$K_I = \sigma\sqrt{\pi a}\, Y$$

Input (8S edition).

$$y_{37}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{37})\Big] z(u), \qquad z_{37}(u) = \begin{bmatrix} c_{37} \\ r_{37} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{37})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{37}\, \psi_{37}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{37} = -\mathcal{L}_\star \psi_{37}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{37} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{37})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(37)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{37}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The coefficient of the $r^{-1/2}$ singularity.

Structure: The near-tip field is a fixed eigenfunction of the elasticity operator in polar coordinates with exponent $-1/2$; K_I is the amplitude of that mode. Fracture mechanics is modal analysis of a singular field.

Reduction: The factor is a modal amplitude.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Structural Engineering.

38. Griffith Fracture Criterion

Origin (LA edition).

$$\sigma_f = \sqrt{\frac{2E\gamma_s}{\pi a}}$$

Input (8S edition).

$$\mathrm{Admit}_{38}(8S) \iff \eta_{\mathrm{ind}}^{(38)} \geq \varepsilon_{\mathrm{ind}} \ \wedge \ \Delta_{8S} > \varepsilon_{\mathrm{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \ \text{ or } \ \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{38}(x,\lambda) = 0, \qquad \partial_x \mathcal{G}_{38} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x \mathcal{G}_{38} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{38} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x,\lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(38)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{38}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An energy balance between released strain energy and created surface.

Structure: Released energy is quadratic in stress (a quadratic form) while surface energy is linear in crack length — the competition of a quadratic against a linear term is what makes the criterion a threshold.

Reduction: The criterion balances a quadratic form against a linear term.

Status: Exact linear in the energy terms.

8S status: Exact linear in the energy terms; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear in the energy terms; carried to the continuum by Lemma D7, and the input was algebraic, so the differential content is entirely supplied by the transform.

39. Energy Release Rate

Origin (LA edition).

$$G = -\frac{\partial \Pi}{\partial a} = \frac{K_I^2}{E'}$$

Input (8S edition).

$$E_{39}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 39}, \qquad \big\|z_{39}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta_{39j}^{(8)} = \sqrt{(c_{39}-c_j)^\top G_5 (c_{39}-c_j) + (r_{39}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{39}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{39} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{39}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{39}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{39} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(39)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{39}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The derivative of a quadratic potential with respect to crack length.

Structure: $\Pi = \frac{1}{2}\mathbf{u}^\top K \mathbf{u} - \mathbf{f}^\top \mathbf{u}$, so G is a sensitivity of a quadratic form to a stiffness change — computable by the virtual crack extension method as $-\frac{1}{2}\mathbf{u}^\top (\partial K/\partial a)\mathbf{u}$.

Reduction: The rate is a derivative of a quadratic form.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input already carried a first-order operator, which the transform leaves in place.

40. Paris Law

Origin (LA edition).

$$\frac{da}{dN} = C(\Delta K)^m$$

Input (8S edition).

$$\log r_{40} = \log r_{\max} + \alpha \log B_5(c_{40}) + \tfrac{1}{3} \log s_{40}, \qquad \log B_5(c_{40}) = \sum_{a=4}^{8} \log b(c_{40,a})$$

Calculus image.

$$d \log r_{40} = \alpha \, d \log B_5(c_{40}) + \tfrac{1}{3} \, d \log s_{40}, \qquad d(d \log r_{40}) = 0, \qquad \left. \frac{\partial \log r_{40}}{\partial \log b(c_{40,a})} \right|_{s_{40}} = \alpha \quad (a = 4, \dots, 8)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T6**, so the calculus image is of class **D6** (exact differential form and elasticity). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the exact differential form and elasticity of Lemma D6, instantiated at the point $c_{40} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D6 needs $B_5(c) > 0$ and $s > 0$ so the logarithms and the form are defined; by Theorem I(4) of the 8S edition this fails exactly at a bump zero or under full suppression, where $r = 0$ and the fibre degenerates. Exactness needs only that $\log r$ be a well-defined C^1 function, which it is on that region.

(Q4) Derivation. Take logarithms of $r = r_{\max} B_5^{\alpha} s^{1/3}$ and of $B_5 = \prod_{a=4}^{8} b(c_a)$, then differentiate. Since the result is the differential of a function, the form is exact; exact forms are closed by $d \circ d = 0$, and their line integrals depend only on the endpoints by the fundamental theorem of calculus for line integrals. The elasticities are read off as the coefficients — as partial derivatives in the factorised coordinates $(\log b(c_4), \dots, \log b(c_8), \log s)$, with the other factors held fixed. If the salience s itself varies over the base, the *total* derivative along c_a carries the extra term $\frac{1}{3}\partial_a \log s$ and is not α; Θ-I(2) is stated in the factorised coordinates precisely to keep that distinction visible.

(Q5) Retraction through the tower. Integrating the form between two lattice points returns the 8S difference of logarithms, $\log r_{40} - \log r_j$, exactly and with no quadrature error, because the form is exact. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission is the first-order condition $\delta \Delta_{8S} = 0$ subject to $\eta_{\text{ind}}^{(40)} \geq \varepsilon_{\text{ind}}$, together with the domain condition $B_5(c_{40}) > 0$ and $s_{40} > 0$ that makes the logarithms and the form defined. Where the domain condition fails, $r_{40} = 0$, the fibre collapses to a point, and ds^2 degenerates to the base metric.

Objects: A power law in the stress intensity range.

Structure: Linear in log-log coordinates, so fitting is ordinary least squares on transformed data. Typically $m \approx 3$ for metals. Integration gives the fatigue life, and the exponent's size is what makes life extremely sensitive to load amplitude.

Reduction: Linear after logarithmic transformation.

Status: Exact linear in logs.

8S status: Exact linear in logs; preserved under the transform by Theorem C and Lemma T6.

Calculus status: Exact linear in logs; carried to the continuum by Lemma D6, and the input already carried a first-order operator, which the transform leaves in place.

Nonlinear and Inelastic Behaviour

41. Hyperelastic Constitutive Relation

Origin (LA edition).

$$S = 2\frac{\partial W}{\partial C}$$

Input (8S edition).

$$E_{41}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node 41}}, \qquad \|z_{41}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta_{41j}^{(8)} = \sqrt{(c_{41}-c_j)^\top G_5 (c_{41}-c_j) + (r_{41}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{41}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{41} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{41}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{41}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{41} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(41)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{41}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Stress as the gradient of a strain energy function.

Structure: Guarantees path independence and thermodynamic consistency. The tangent stiffness $\partial S/\partial E$ is the Hessian of W, symmetric by construction — which is why hyperelastic finite element tangents are always symmetric.

Reduction: Stress is a gradient; tangent stiffness is a Hessian.

Status: Linearised at the tangent level.

8S status: Linearised at the tangent level; the 8S image inherits the same approximation status (Theorem C, Lemma T3).

Calculus status: Linearised at the tangent level; carried to the continuum by Lemma D3, and the input already carried a first-order operator, which the transform leaves in place.

42. Neo-Hookean Model

Origin (LA edition).

$$W = \frac{\mu}{2}(I_1 - 3)$$

Input (8S edition).

$$E_{42}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 42}, \qquad \big\|z_{42}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta_{42j}^{(8)} = \sqrt{(c_{42}-c_j)^\top G_5 (c_{42}-c_j) + (r_{42}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{42}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{42} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{42}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{42}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{42} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u\cdot du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r\cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(42)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{42}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Energy depending on the first invariant $\operatorname{tr} C$.

Structure: The simplest hyperelastic model, reducing to linear elasticity at small strain. Since W is linear in $\operatorname{tr} C$, the stress is affine in C — but C is quadratic in F, so the overall response is nonlinear.

Reduction: Linear in the invariant, quadratic in F.

Status: Linearised at small strain.

8S status: Linearised at small strain; the 8S image inherits the same approximation status (Theorem C, Lemma T3).

Calculus status: Linearised at small strain; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

43. Mooney–Rivlin Model

Origin (LA edition).

$$W = C_{10}(I_1 - 3) + C_{01}(I_2 - 3)$$

Input (8S edition).

$$E_{43}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 43}, \qquad \|z_{43}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta_{43j}^{(8)} = \sqrt{(c_{43} - c_j)^\top G_5 (c_{43} - c_j) + (r_{43} - r_j)^2}$$

Calculus image.

$$\mathcal{E}_{43}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{43} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{43}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{43}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{43} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star\phi, \phi\rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c, u) = (c, r(c)u)$, use $u{\cdot}du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r{\cdot}dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(43)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{43}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A two-invariant energy function.

Structure: Adding the second invariant improves the fit at moderate strain. Fitting C_{10}, C_{01} to test data is a linear least-squares problem, since W is linear in the parameters even though it is nonlinear in the deformation.

Reduction: Linear in the material parameters.

Status: Exact linear in the parameters.

8S status: Exact linear in the parameters; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear in the parameters; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

44. Ogden Model

Origin (LA edition).

$$W = \sum_p \frac{\mu_p}{\alpha_p}(\lambda_1^{\alpha_p} + \lambda_2^{\alpha_p} + \lambda_3^{\alpha_p} - 3)$$

Input (8S edition).

$$P_5^{(44)} = C_5 C_5^{+} = P_4 + \frac{w_{44} w_{44}^{\top}}{\|w_{44}\|_2^2}, \qquad w_{44} = (I - C_4 C_4^{+}) e_{44}, \qquad \eta_{\text{ind}}^{(44)} = \frac{\|w_{44}\|_2^2}{\|e_{44}\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_{44} - P_{\mathcal{V}} e_{44},\ v \rangle = 0 \ \ \forall v \in \mathcal{V}, \qquad \eta_{\text{ind}}^{(44)} = \frac{\det \mathrm{Gr}(C_4, e_{44})}{\det \mathrm{Gr}(C_4)(\|e_{44}\|^2 + \varepsilon)}, \qquad \nabla_{e_{44}} \|w_{44}\|^2 = 2\, w_{44}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_{44} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_{44}\|^2/(\|e_{44}\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(44)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{44}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Energy written directly in the principal stretches — the eigenvalues of U.

Structure: Linear in μ_p but nonlinear in α_p, so calibration is a separable nonlinear least-squares problem: solve linearly for μ_p given α_p, and search over the α_p. That separability is exactly the variable projection method.

Reduction: Linear in half the parameters.

Status: Exact linear in the μ_p.

8S status: Exact linear in the μ_p; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear in the μ_p; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

45. Von Mises Yield Criterion

Origin (LA edition).

$$\sqrt{\tfrac{3}{2}\sigma' : \sigma'} = \sigma_y$$

Input (8S edition).

$$\max_{n_{45} \in \mathbb{Z}^5} \ \Delta_{8S} = \mathrm{Score}(\mathcal{M}_8) - \mathrm{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{45}\|_1 \leq M, \ \ \eta_{\text{ind}}^{(45)} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{45}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(45)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_{45}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(45)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{45} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(45)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{45}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A norm ball in deviatoric stress space.

Structure: The yield surface is a cylinder of radius $\sqrt{2/3}\,\sigma_y$ about the hydrostatic axis — smooth and convex, which makes the flow rule single-valued everywhere. Convexity of the yield surface follows from Drucker's postulate.

Reduction: Yield is a norm constraint.

Status: Exact linear (norm constraint).

8S status: Exact linear (norm constraint); preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear (norm constraint); carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

46. Tresca Yield Criterion

Origin (LA edition).

$$\max_{i,j} |\sigma_i - \sigma_j| = \sigma_y$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(46)}_k = \mu^{(46)}_k \phi^{(46)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(46)}_1 \leq \mu^{(46)}_2 \leq \cdots \leq \mu^{(46)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(46)}_k) = \mu^{(46)}_k \phi^{(46)}_k, \qquad \mu^{(46)}_2 = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{46} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(46)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{46}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A max over eigenvalue differences.

Structure: The yield surface is a hexagonal prism — a polyhedron, hence piecewise linear. Corners make the normal non-unique, so the flow rule is set-valued there. Piecewise linearity means Tresca plasticity can be posed as a linear complementarity problem.

Reduction: Yield is a polyhedral constraint on eigenvalues.

Status: Exact linear (piecewise).

8S status: Exact linear (piecewise); preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear (piecewise); carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

47. Mohr–Coulomb Criterion

Origin (LA edition).

$$\tau = c - \sigma_n \tan\varphi$$

Input (8S edition).

$$\left(g^{(5)}_{47j},\ \delta^{(8)}_{47j},\ g^{(3)}_{47j},\ g^{(\mathrm{J})}_{47j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{47j}, \qquad g^{(3)}_{47j} \leq g^{(5)}_{47j}$$

Calculus image.

$$d_g(z_{47}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{47j} \leq d_g(z_{47}, z_j), \qquad g^{(3)}_{47j} \leq g^{(5)}_{47j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{47} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \leq d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{47j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(47)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{47}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A pressure-dependent linear criterion.

Structure: Linear in the normal stress, giving an irregular hexagonal pyramid in principal stress space. Appropriate for soils and rock, where hydrostatic pressure strengthens the material — the pressure dependence is exactly what von Mises omits.

Reduction: The criterion is linear in the stress components.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

48. Drucker–Prager Criterion

Origin (LA edition).

$$\sqrt{J_2} = A + BI_1$$

Input (8S edition).

$$\max_{n_{48} \in \mathbb{Z}^5} \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{48}\|_1 \leq M, \;\; \eta^{(48)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{48}\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(48)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_{48}\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(48)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{48} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(48)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{48}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A smooth cone in stress space.

Structure: A smooth approximation to Mohr–Coulomb: a second-order cone constraint, hence convex and free of corners. Being an SOC constraint, plasticity with this criterion is a conic programme.

Reduction: The criterion is a second-order cone constraint.

Status: Exact linear (conic).

8S status: Exact linear (conic); preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear (conic); carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

49. Associated Flow Rule

Origin (LA edition).

$$\dot{\varepsilon}^p = \dot{\lambda} \frac{\partial f}{\partial \sigma}$$

Input (8S edition).

$$\left(g^{(5)}_{49j},\ \delta^{(8)}_{49j},\ g^{(3)}_{49j},\ g^{(\mathrm{J})}_{49j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{49j}, \qquad g^{(3)}_{49j} \leq g^{(5)}_{49j}$$

Calculus image.

$$d_g(z_{49}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{49j} \leq d_g(z_{49}, z_j), \qquad g^{(3)}_{49j} \leq g^{(5)}_{49j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{49} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{49j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(49)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{49}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Plastic strain rate normal to the yield surface.

Structure: Normality plus convexity gives maximum plastic dissipation, and it makes the elastoplastic tangent stiffness symmetric — a major computational advantage. Non-associated rules (common in soils) break that symmetry and force unsymmetric solvers.

Reduction: The rule is a normal-cone condition.

Status: Exact linear as a normality condition.

8S status: Exact linear as a normality condition; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear as a normality condition; carried to the continuum by Lemma D9, and the input already carried a first-order operator, which the transform leaves in place.

50. Isotropic Hardening

Origin (LA edition).

$$\sigma_y = \sigma_{y0} + H\bar{\varepsilon}^p$$

Input (8S edition).

$$y_{50}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{50})\Big] z(u), \qquad z_{50}(u) = \begin{bmatrix} c_{50} \\ r_{50} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{50})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{50}\, \psi_{50}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{50} = -\mathcal{L}_\star \psi_{50}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{50} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs

(R1) and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{50})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2, K_3, J_2, J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(50)})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{50}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Uniform expansion of the yield surface.

Structure: Scales the yield surface without moving its centre, so it is a scalar multiple of the original set. Cannot represent the Bauschinger effect, since the surface stays centred — that requires the next entry.

Reduction: Hardening is a scalar scaling of the yield set.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

51. Kinematic Hardening

Origin (LA edition).

$$f = \|\sigma' - \alpha\| - \sigma_y$$

Input (8S edition).

$$y_{51}(\tau,u) = \Pi(\tau)\Big[(\Gamma W\Gamma)\otimes \mathsf{B}(\mathsf{A}_{51})\Big]z(u), \qquad z_{51}(u) = \begin{bmatrix} c_{51} \\ r_{51}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{51})(c) = \int_{\mathcal{C}} \gamma(c)\,W_\sigma(c,c')\,\gamma(c')\,\mathsf{A}_{51}\,\psi_{51}(c')\,p(c')\,dc', \qquad \partial_\tau\psi_{51} = -\mathcal{L}_\star\psi_{51}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{51}\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{51})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(51)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{51}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Translation of the yield surface by a back-stress tensor α.

Structure: A rigid translation in stress space — an affine shift of the yield set. Captures the Bauschinger effect and cyclic behaviour. Combined isotropic–kinematic models scale and translate simultaneously.

Reduction: Hardening is an affine shift of the yield set.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

52. Ramberg–Osgood Relation

Origin (LA edition).

$$\varepsilon = \frac{\sigma}{E} + K \left(\frac{\sigma}{E} \right)^n$$

Input (8S edition).

$$\left. \frac{\partial \dot{\theta}}{\partial \theta} \right|_{\theta^\star} = -\mathcal{L}_{8S}, \qquad \left. \frac{\partial \Phi}{\partial p} \right|_{p^\star} = \Sigma'(2J_2 W + 4J_3 \widehat{H}_{p^\star}), \qquad \Sigma' = \operatorname{diag}(p_{52}^\star(1 - p_{52}^\star))$$

Calculus image.

$$\mathcal{N}_{52}(x_0 + \xi) = \mathcal{N}_{52}(x_0) + D\mathcal{N}_{52}[x_0]\, \xi + R_{52}[\xi], \qquad \big\| R_{52}[\xi] \big\| \leq \tfrac{1}{2} \sup_{\|\zeta\| \leq \|\xi\|} \big\| D^2 \mathcal{N}_{52}[x_0 + \zeta] \big\|\, \|\xi\|^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T13**, so the calculus image is of class **D13** (Fréchet derivative with a quantified remainder). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fréchet derivative with a quantified remainder of Lemma D13, instantiated at the point $c_{52} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D13 needs $\mathcal{N}$ twice Fréchet differentiable on a neighbourhood of x_0 for Taylor's theorem with integral remainder, and needs condition **(C)**, $\frac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} < 1$, for the point x_0 to exist and be unique — by Θ-H(2) that is exactly strict convexity of the free energy. Beyond (C) the expansion point is not unique and the remainder bound is void.

(Q4) Derivation. Expand $\mathcal{N}$ about the fixed point by Taylor's theorem with integral remainder, $R[\xi] = \int_0^1 (1 - t) D^2 \mathcal{N}[x_0 + t\xi][\xi, \xi]\, dt$, and bound the integral by the supremum of the second derivative. The phase block of

$D\mathcal{N}$ is $-\mathcal{L}_\star$ by Θ-G; the gate block is $\nabla^2\mathcal{F}$, whose positive definiteness under (C) is Θ-H(2) and which certifies that x_0 is the unique expansion point.

(Q5) Retraction through the tower. Discretising both Jacobian blocks returns the 8S two-block Jacobian, $-\mathcal{L}_{8S}$ and $\Sigma'(2J_2W+4J_3\widehat{H})$; the remainder term has no 8S counterpart and is new information supplied by the calculus edition. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(52)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{52}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A linear elastic term plus a power-law plastic term.

Structure: Explicitly the sum of a linear and a nonlinear contribution — the decomposition of total strain into elastic and plastic parts made concrete. Only the first term admits a matrix form.

Reduction: Linear in the elastic part only.

Status: Linearised in the elastic regime.

8S status: Linearised about the 8S fixed point $(p^\star, \theta^\star)$ (Lemma T13); the image is an approximation exactly as the input is.

Calculus status: Linearised, now with a quantified remainder: $\|R\| \leq \frac{1}{2}\sup\|D^2\mathcal{N}\|\,\|\xi\|^2$ (Lemma D13). The calculus edition supplies a modulus of validity the 8S edition could only assert.

53. Maxwell Viscoelastic Model

Origin (LA edition).

$$\dot{\varepsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\eta}$$

Input (8S edition).

$$y_{53}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{53})\Big] z(u), \qquad z_{53}(u) = \begin{bmatrix} c_{53} \\ r_{53}u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{53})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{53}\, \psi_{53}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{53} = -\mathcal{L}_\star \psi_{53}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{53} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind.

Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{53})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(53)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{53}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A first-order linear ODE — spring and dashpot in series.

Structure: Relaxation modulus $E(t) = Ee^{-t/\tau}$ with $\tau = \eta/E$: the response is a matrix exponential. Fluid-like, since stress relaxes to zero under constant strain. Generalised Maxwell models sum several such exponentials — a Prony series.

Reduction: The model is a linear ODE, hence a matrix exponential.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input already carried a first-order operator, which the transform leaves in place.

54. Kelvin–Voigt Model

Origin (LA edition).

$$\sigma = E\varepsilon + \eta\dot{\varepsilon}$$

Input (8S edition).

$$y_{54}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{54})\Big] z(u), \qquad z_{54}(u) = \begin{bmatrix} c_{54} \\ r_{54} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{54})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{54}\, \psi_{54}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{54} = -\mathcal{L}_\star \psi_{54}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{54} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind.

Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{54})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(54)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{54}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Spring and dashpot in parallel; also linear.

Structure: Creep compliance $J(t) = \frac{1}{E}(1 - e^{-t/\tau})$ — solid-like, since strain saturates. Maxwell and Kelvin–Voigt are the two elementary two-element models, and both are single-pole linear systems.

Reduction: The model is a linear ODE.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input already carried a first-order operator, which the transform leaves in place.

55. Standard Linear Solid

Origin (LA edition).

$$\sigma + \tau_\sigma \dot{\sigma} = E_R(\varepsilon + \tau_\varepsilon \dot{\varepsilon})$$

Input (8S edition).

$$y_{55}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{55})\Big] z(u), \qquad z_{55}(u) = \begin{bmatrix} c_{55} \\ r_{55} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{55})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{55}\, \psi_{55}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{55} = -\mathcal{L}_\star \psi_{55}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{55} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the

Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{55})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(55)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{55}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A three-parameter linear model with two time constants.

Structure: Exhibits both instantaneous elasticity and equilibrium stiffness, so it captures creep and relaxation together. As a linear system it is a first-order transfer function with one pole and one zero.

Reduction: The model is a first-order linear system.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input already carried a first-order operator, which the transform leaves in place.

56. Boltzmann Superposition Integral

Origin (LA edition).

$$\sigma(t) = \int_{-\infty}^{t} E(t-s)\dot{\varepsilon}(s)\, ds$$

Input (8S edition).

$$\widehat{W}^{(56)}(k) = \sum_{m} W(m)\, e^{-2\pi i \langle k,m\rangle/L}, \qquad F_5 = F^{\otimes 5}, \qquad F_5 W F_5^{*} = \mathrm{diag}(\widehat{W}(k))$$

Calculus image.

$$\widehat{W_\sigma * \psi_{56}}(\xi) = \widehat{W_\sigma}(\xi)\, \widehat{\psi_{56}}(\xi), \qquad \widehat{W_\sigma}(\xi) = e^{-\sigma_c^2 \xi^{\top} G_5^{-1} \xi/2}, \qquad \widehat{\partial_a \psi_{56}}(\xi) = i\, \xi_a\, \widehat{\psi_{56}}(\xi)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T12**, so the calculus image is of class **D12** (Fourier transform and pseudodifferential symbol). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fourier transform and pseudodifferential symbol of Lemma D12, instantiated at the point $c_{56} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D12 needs $W_\sigma \in L^1$ for the convolution theorem, which the Gaussian satisfies, and ψ in a space where the transform is defined; $L^1 \cap L^2$ suffices and is standing. Translation invariance of the mask is required for W_σ to act as a convolution at all.

(Q4) Derivation. Apply the convolution theorem, then evaluate the transform of the Gaussian by completing the square in the exponent: $\int e^{-x^{\top} A x} e^{-i\xi\cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^{\top} A^{-1} \xi/4}$ with $A = G_5/2\sigma_c^2$. The differentiation rule follows by integrating by parts in the transform integral, the boundary term vanishing by decay. Together these exhibit the coupling as a pseudodifferential operator with Gaussian symbol.

(Q5) Retraction through the tower. On the lattice torus the Fourier transform becomes the five-fold DFT $F_5 = F^{\otimes 5}$ and the symbol becomes the eigenvalue $\widehat{W}(k)$ of the 8S circulant. Composing with the 8S retraction of Theorem

J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(56)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{56}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Convolution with the relaxation modulus.

Structure: Linearity is the entire content — the response to a sum of histories is the sum of responses. Discretised, it is multiplication by a lower-triangular Toeplitz matrix, so viscoelasticity is a causal linear filter.

Reduction: The response is a Toeplitz matrix product.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T12.

Calculus status: Exact linear; carried to the continuum by Lemma D12, and the input already carried a first-order operator, which the transform leaves in place.

57. Complex Modulus

Origin (LA edition).

$$E^*(\omega) = E'(\omega) + iE''(\omega)$$

Input (8S edition).

$$\widehat{W}^{(57)}(k) = \sum_m W(m)\, e^{-2\pi i\langle k,m\rangle/L}, \qquad F_5 = F^{\otimes 5}, \qquad F_5 W F_5^* = \operatorname{diag}(\widehat{W}(k))$$

Calculus image.

$$\widehat{W_\sigma * \psi_{57}}(\xi) = \widehat{W_\sigma}(\xi)\,\widehat{\psi_{57}}(\xi), \qquad \widehat{W_\sigma}(\xi) = e^{-\sigma_c^2 \xi^\top G_5^{-1}\xi/2}, \qquad \widehat{\partial_a \psi_{57}}(\xi) = i\,\xi_a\,\widehat{\psi_{57}}(\xi)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T12**, so the calculus image is of class **D12** (Fourier transform and pseudodifferential symbol). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Fourier transform and pseudodifferential symbol of Lemma D12, instantiated at the point $c_{57} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D12 needs $W_\sigma \in L^1$ for the convolution theorem, which the Gaussian satisfies, and ψ in a space where the transform is defined; $L^1 \cap L^2$ suffices and is standing. Translation invariance of the mask is required for W_σ to act as a convolution at all.

(Q4) Derivation. Apply the convolution theorem, then evaluate the transform of the Gaussian by completing the square in the exponent: $\int e^{-x^\top Ax} e^{-i\xi\cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1}\xi/4}$ with $A = G_5/2\sigma_c^2$. The differentiation rule follows by integrating by parts in the transform integral, the boundary term vanishing by decay. Together these exhibit the coupling as a pseudodifferential operator with Gaussian symbol.

(Q5) Retraction through the tower. On the lattice torus the Fourier transform becomes the five-fold DFT $F_5 = F^{\otimes 5}$ and the symbol becomes the eigenvalue $\widehat{W}(k)$ of the 8S circulant. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(57)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{57}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The Fourier transform of the relaxation modulus.

Structure: Storage and loss moduli are the real and imaginary parts, related by Kramers–Kronig — that is, by the Hilbert transform matrix. Causality forces one to determine the other, exactly as in the optics case.

Reduction: The modulus is a Fourier-domain multiplier.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T12.

Calculus status: Exact linear; carried to the continuum by Lemma D12, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

58. Time–Temperature Superposition (WLF)

Origin (LA edition).

$$\log a_T = \frac{-C_1(T - T_0)}{C_2 + T - T_0}$$

Input (8S edition).

$$y_{58}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{58})\Big] z(u), \qquad z_{58}(u) = \begin{bmatrix} c_{58} \\ r_{58} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{58})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{58}\, \psi_{58}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{58} = -\mathcal{L}_\star \psi_{58}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{58} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{58})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(58)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{58}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A horizontal shift of curves on a logarithmic time axis.

Structure: Shifting is a translation in $\log t$, hence a linear operation in that coordinate; master curve construction is fitting a single curve to shifted data, which is a linear least-squares problem given the shift factors.

Reduction: Superposition is translation in log time.

Status: Exact linear in log time.

8S status: Exact linear in log time; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear in log time; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

59. Norton Creep Law

Origin (LA edition).

$$\dot{\varepsilon}_{ss} = A\sigma^n e^{-Q/RT}$$

Input (8S edition).

$$\log r_{59} = \log r_{\max} + \alpha \log B_5(c_{59}) + \tfrac{1}{3}\log s_{59}, \qquad \log B_5(c_{59}) = \sum_{a=4}^{8} \log b(c_{59,a})$$

Calculus image.

$$d\log r_{59} = \alpha\, d\log B_5(c_{59}) + \tfrac{1}{3}\, d\log s_{59}, \qquad d(d\log r_{59}) = 0, \qquad \left.\frac{\partial \log r_{59}}{\partial \log b(c_{59,a})}\right|_{s_{59}} = \alpha \quad (a = 4, \dots, 8)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T6**, so the calculus image is of class **D6** (exact differential form and elasticity). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the exact differential form and elasticity of Lemma D6, instantiated at the point $c_{59} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D6 needs $B_5(c) > 0$ and $s > 0$ so the logarithms and the form are defined; by Theorem I(4) of the 8S edition this fails exactly at a bump zero or under full suppression, where $r = 0$ and the fibre degenerates. Exactness needs only that $\log r$ be a well-defined C^1 function, which it is on that region.

(Q4) Derivation. Take logarithms of $r = r_{\max} B_5^{\alpha} s^{1/3}$ and of $B_5 = \prod_{a=4}^{8} b(c_a)$, then differentiate. Since the result is the differential of a function, the form is exact; exact forms are closed by $d \circ d = 0$, and their line integrals depend only on the endpoints by the fundamental theorem of calculus for line integrals. The elasticities are read off as the coefficients — as partial derivatives in the factorised coordinates $(\log b(c_4), \dots, \log b(c_8), \log s)$, with the other factors held fixed. If the salience s itself varies over the base, the *total* derivative along c_a carries the extra term $\frac{1}{3}\partial_a \log s$ and is not α; Θ-I(2) is stated in the factorised coordinates precisely to keep that distinction visible.

(Q5) Retraction through the tower. Integrating the form between two lattice points returns the 8S difference of logarithms, $\log r_{59} - \log r_j$, exactly and with no quadrature error, because the form is exact. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission is the first-order condition $\delta\Delta_{8S} = 0$ subject to $\eta_{\text{ind}}^{(59)} \geq \varepsilon_{\text{ind}}$, together with the domain condition $B_5(c_{59}) > 0$ and $s_{59} > 0$ that makes the logarithms and the form defined. Where the domain condition fails, $r_{59} = 0$, the fibre collapses to a point, and ds^2 degenerates to the base metric.

Objects: A power law in stress with Arrhenius temperature dependence.

Structure: Linear in $\ln\dot{\varepsilon}$ against $\ln\sigma$ and $1/T$, so parameter fitting is a two-variable linear regression. Typically $n = 3$–8, and the exponent identifies the dominant creep mechanism.

Reduction: Linear after logarithmic transformation.

Status: Exact linear in logs.

8S status: Exact linear in logs; preserved under the transform by Theorem C and Lemma T6.

Calculus status: Exact linear in logs; carried to the continuum by Lemma D6, and the input already carried a first-order operator, which the transform leaves in place.

60. Larson–Miller Parameter

Origin (LA edition).

$$P = T(C + \log t_r)$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta_{\mathrm{ind}}^{(60)} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind}\mathcal{L}_\star = 0, \qquad \det \mathrm{Gr}(C_4, e_{60}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{60} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker\mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker\mathcal{L}_\star$ becomes $\dim\ker\mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(60)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{60}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A time–temperature equivalence for rupture.

Structure: Collapses rupture data at many temperatures onto a single master curve, which is a rank-one factorisation of the two-variable rupture surface: $\log t_r$ becomes a function of P alone. Data collapse is a rank-one approximation.

Reduction: The collapse is a rank-one factorisation.

Status: Exact linear as a rank-one collapse.

8S status: Exact linear as a rank-one collapse; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear as a rank-one collapse; carried to the continuum by Lemma D5, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Mechanical Engineering.